

Expanded Search for Toroidal Isomers

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Preliminary Exam / Masters Defense

2026-05-13

Title

Liquid Drop Model (1970s) - Original Theory

Spherical Nucleus Binding Energy

$$E_B = a_V A - a_S A^{2/3} - a_C \frac{Z(Z-1)}{A^{1/3}} - a_A \frac{(N-Z)^2}{A} \pm \delta(N, Z)$$

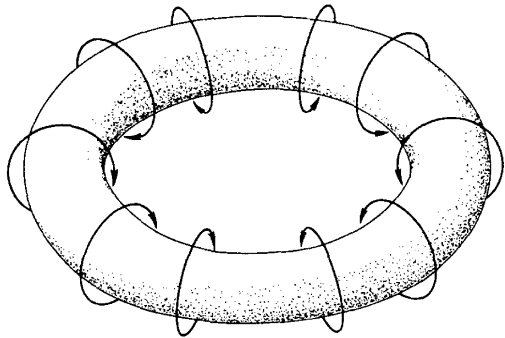
Liquid Drop Model (1970s) - Original Theory

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- E_B depends on **spherical** assumptions.

What happens under deformed Toroidal configurations at high angular momentum?



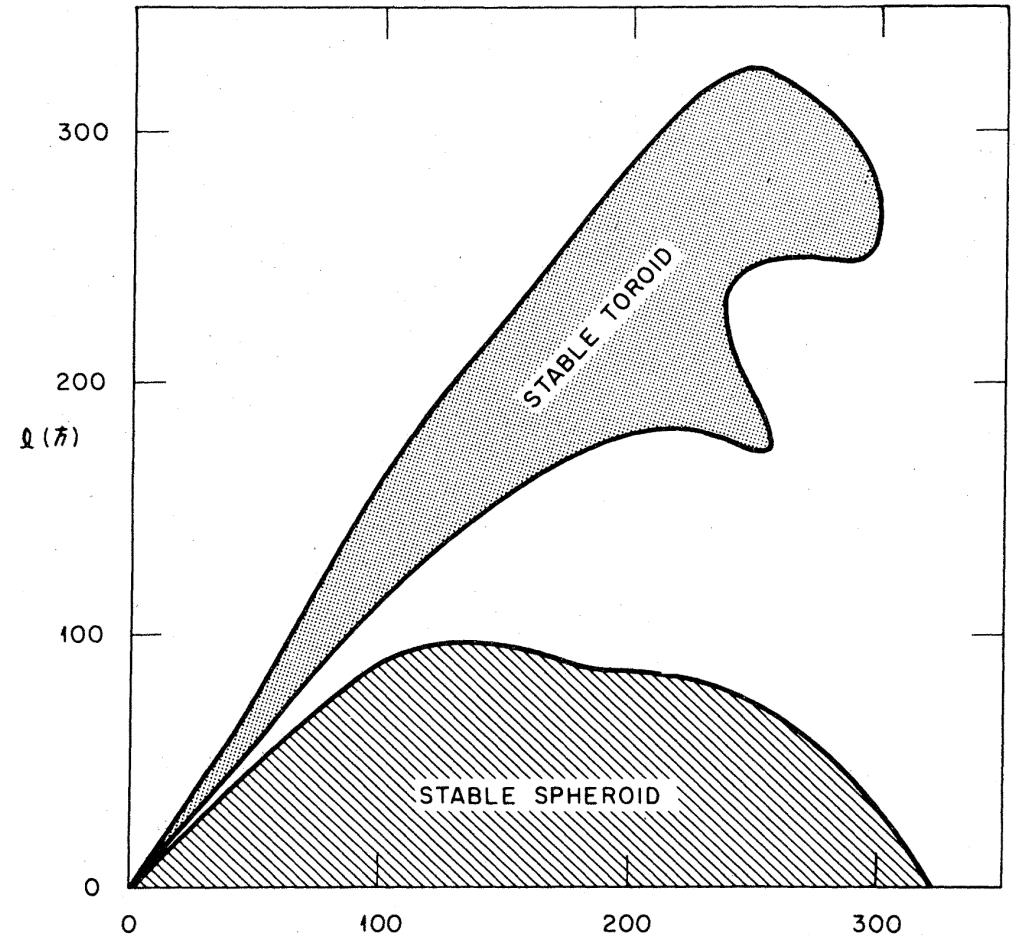
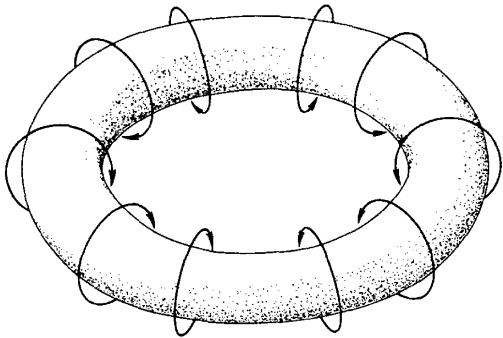
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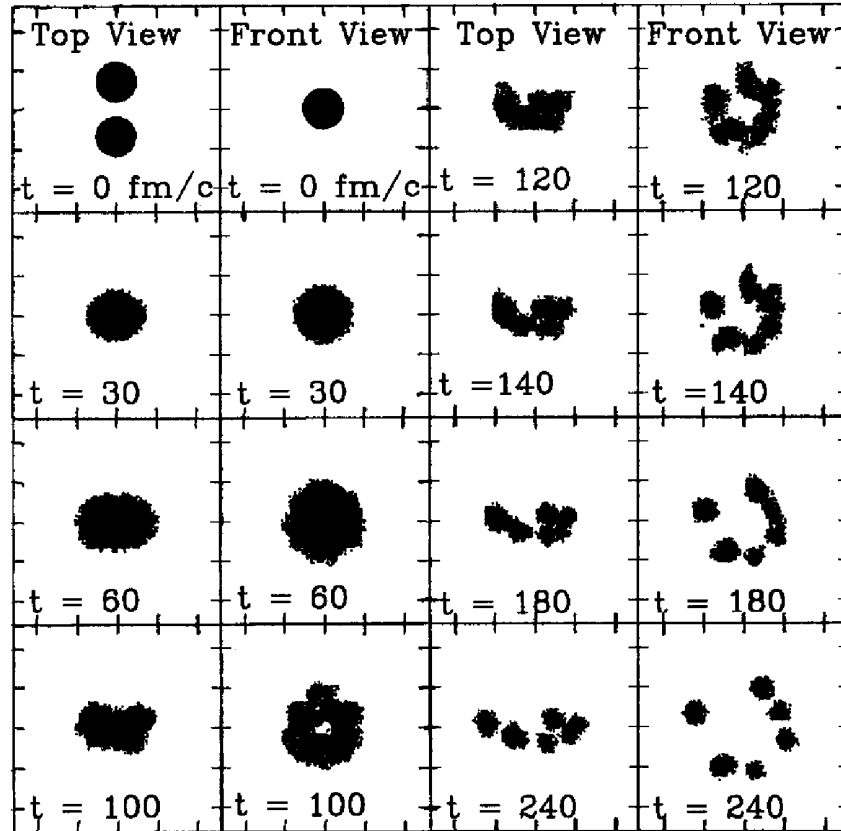
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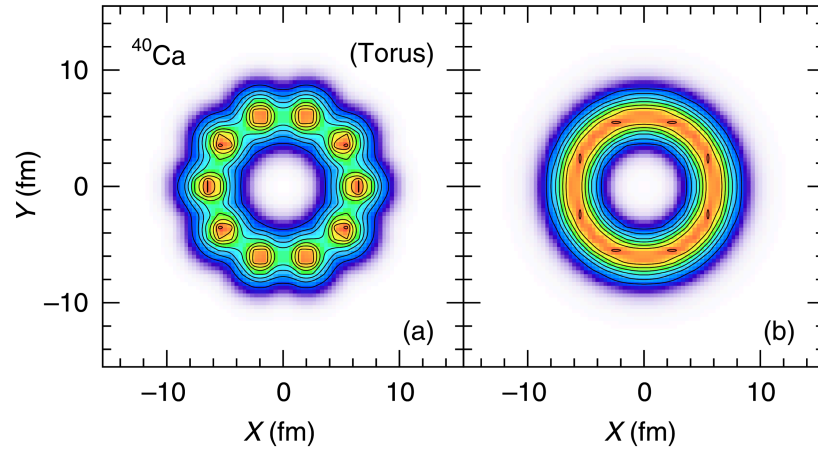
Boltzmann-Uehling-Uhlenbeck (BUU) (1990s) - Dynamic Formation

$^{92}\text{Mo} + ^{92}\text{Mo}$, $E/A = 100$ MeV, $b=0$, Stiff EOS



- Direct collisions produced outwardly expanding toroids.
- Toroids seem to be metastabilized
- Decay into symmetric particles
 - $r_{\text{particle}} \approx d_{\text{toroid}}$

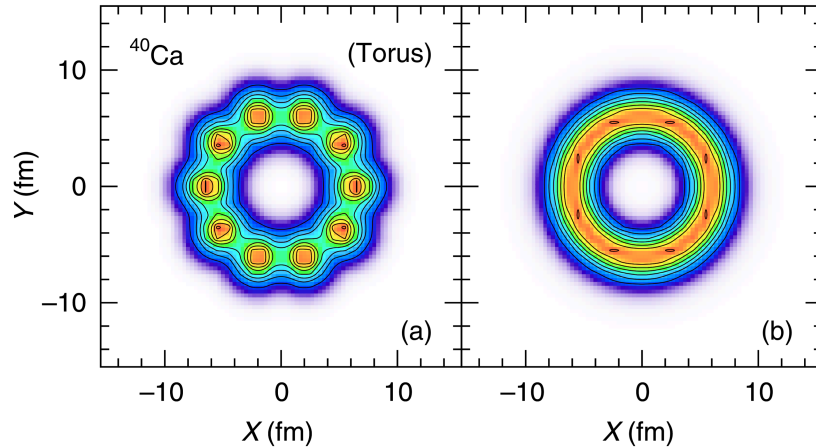
Cranked Skyrme Hartree Fock (cSHF) (2010s) - Specific Energy Predictions



PRL **109**, 232503 (2012)

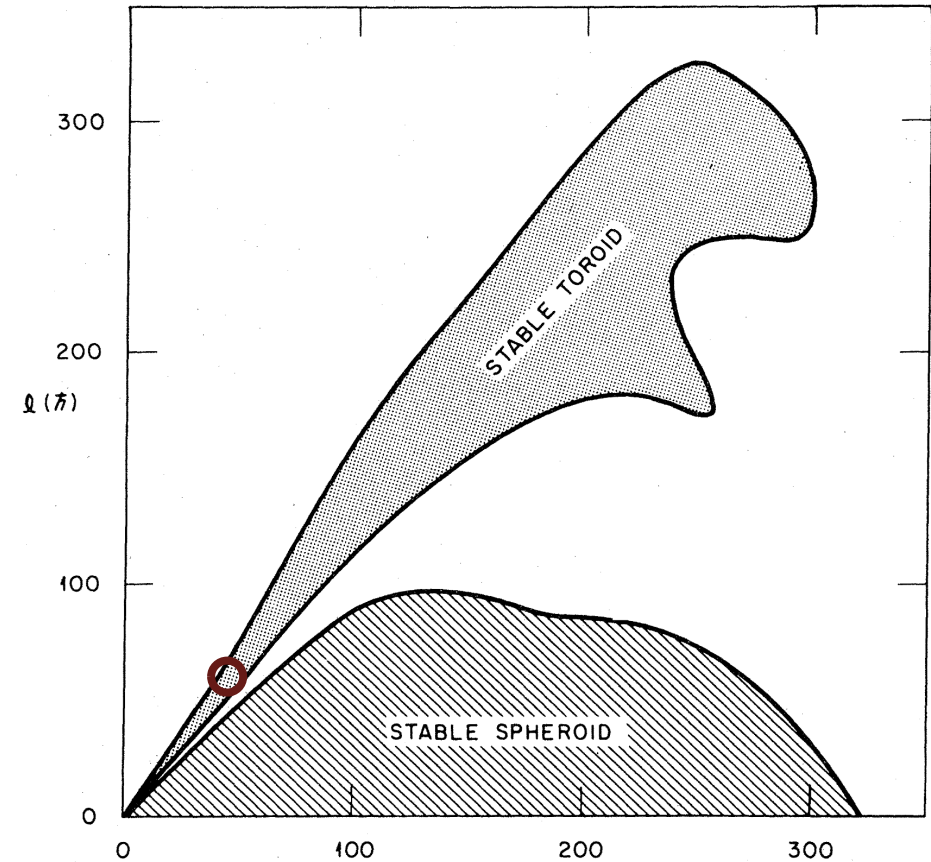
- Predicted state in ^{40}Ca
 - $J_z = 60\hbar$
 - $E^* \approx 175 \text{ MeV}$
- Angular momentum and mass consistent with Wong's LDM

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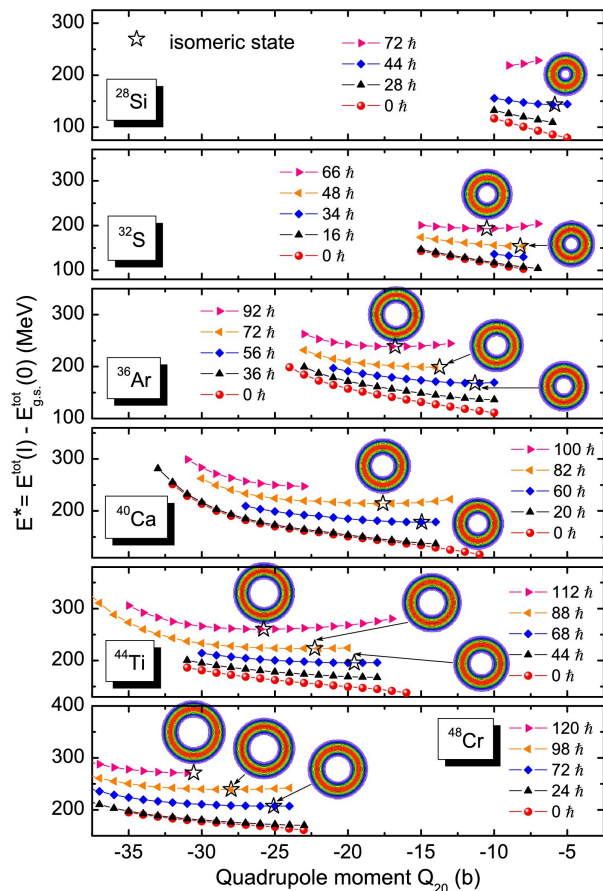
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PLB **738**, 401-404 (2014)

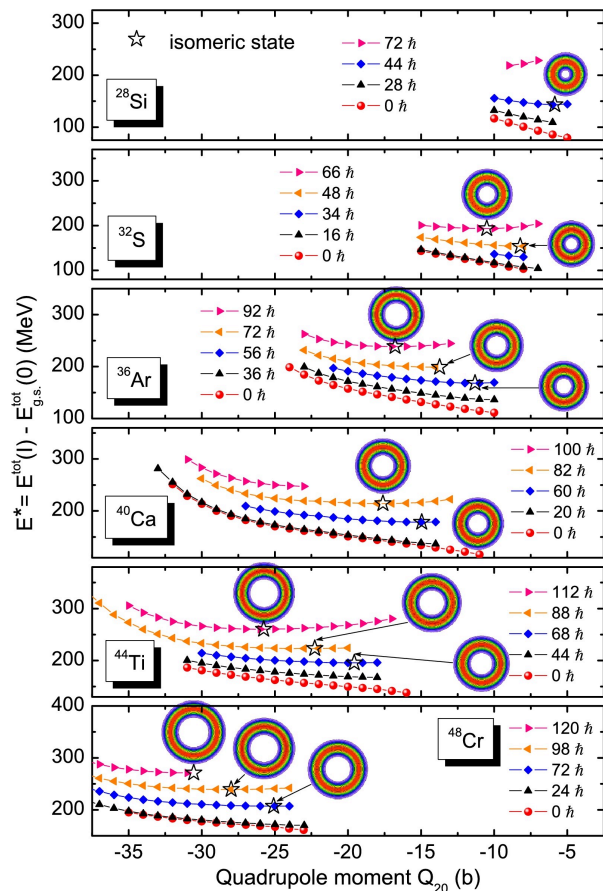


A. Staszczak and C.-Y. Wong predict 18 Toroidal Isomers

	E^* [MeV]	I [$\hbar$]	d [fm]	ρ_{max}/ρ_0
^{28}Si	143.18	44	1.45	0.74
^{32}S	153.87	48	1.42	0.76
	193.35	66	1.40	0.67
^{36}Ar	168.03	56	1.40	0.78
	198.63	72	1.39	0.71
	238.56	92	1.37	0.64
^{40}Ca	178.36	60	1.40	0.79
	214.23	82	1.39	0.73
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PLB **738**, 401-404 (2014)



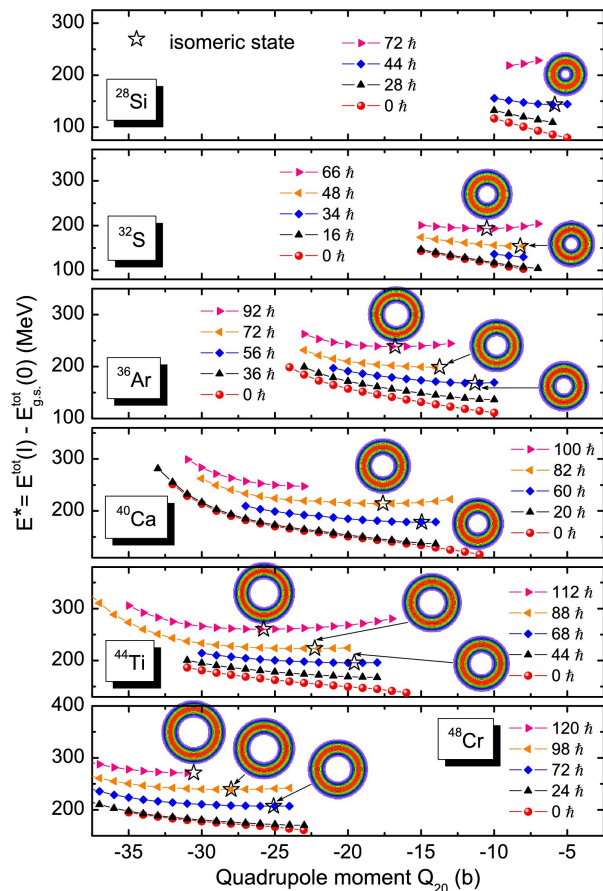
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Ichikawa's ^{40}Ca is reproduced

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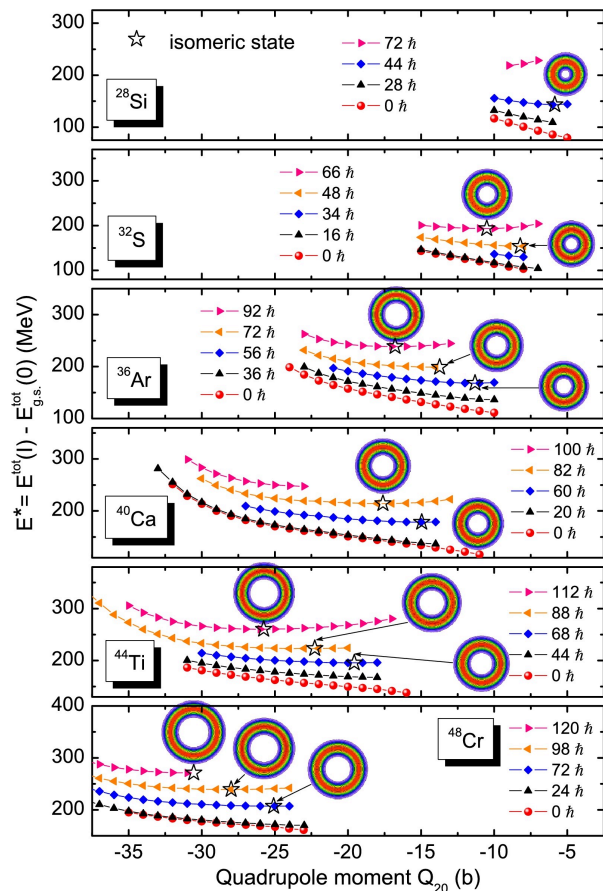
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Minor axis stays about constant ($d \approx 1.4$ fm)

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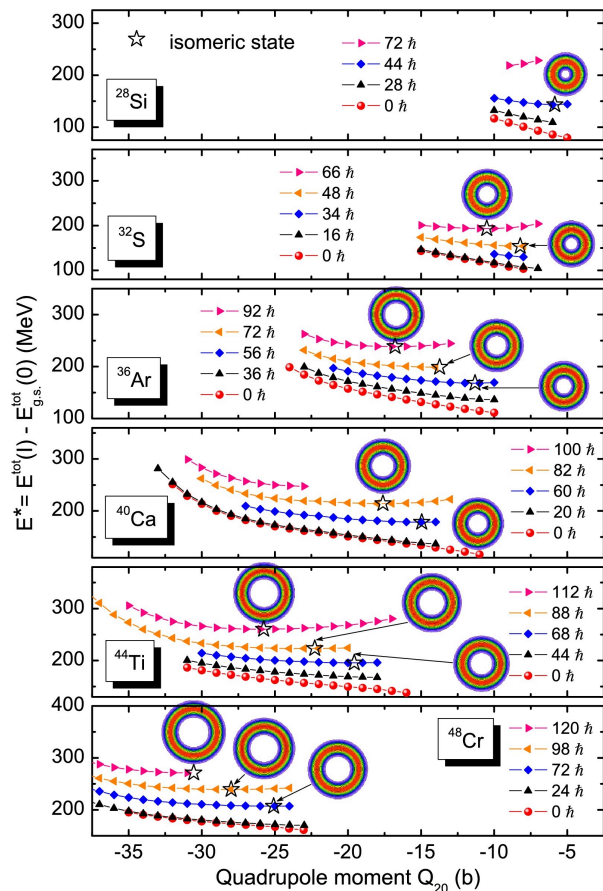
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Low density

Cranked Skyrme Hartree Fock (cSHF) (2010s) - Specific Energy Predictions

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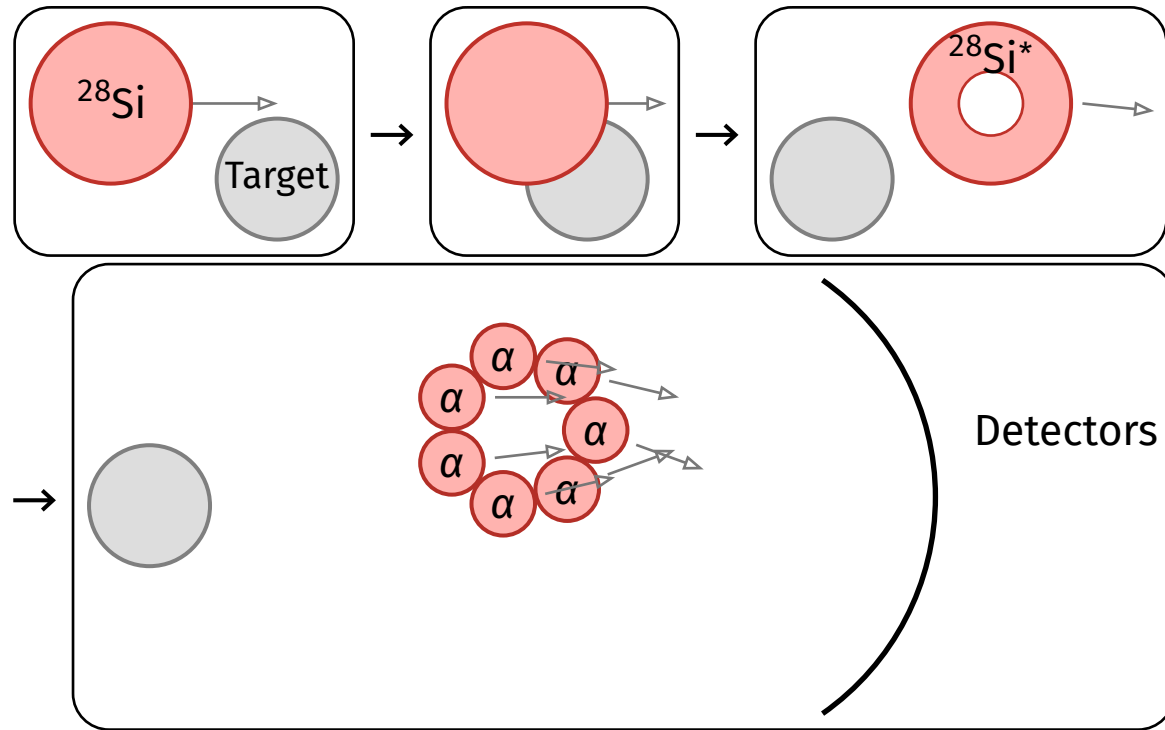


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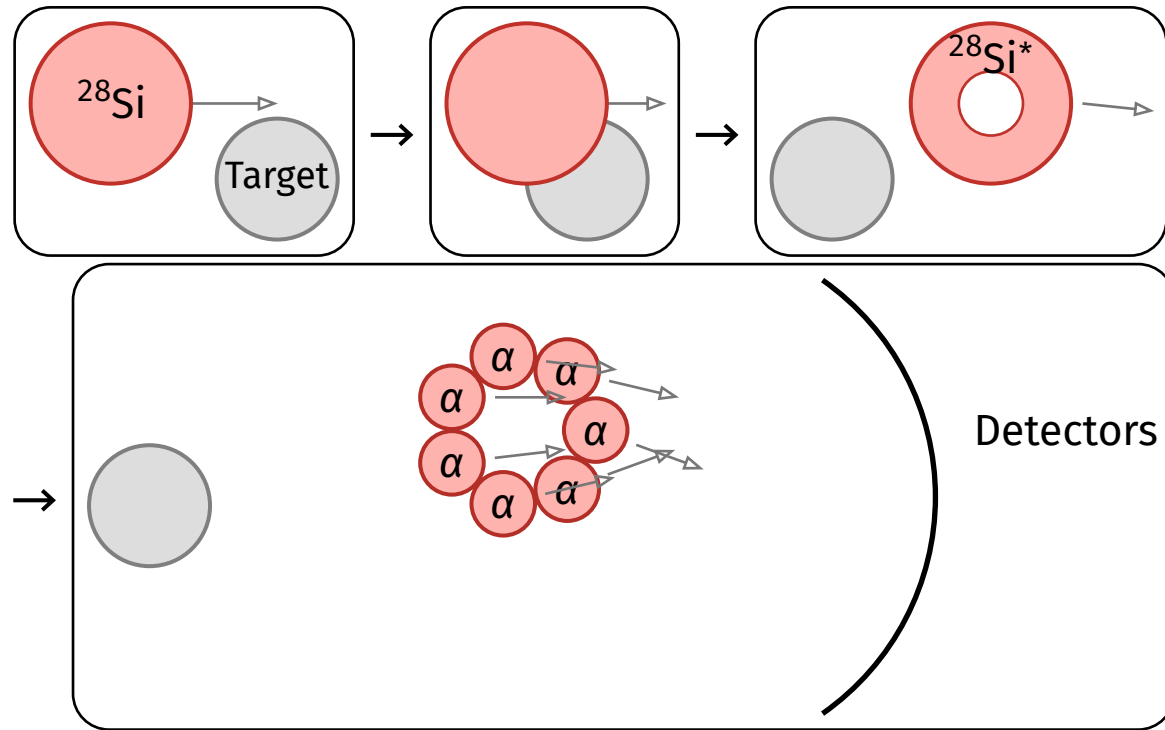
Several states to investigate, including $^{28}\text{Si}(E^* = 143 \text{ MeV})$

First Experimental Evidence (2019)



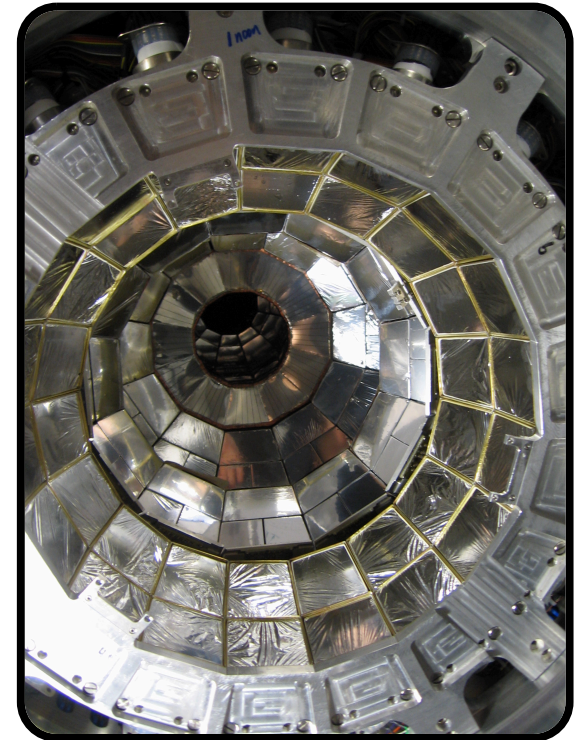
$$E_{^{28}\text{Si}}^* = \underbrace{\sum_i T_{\alpha_i}}_{E_{\text{rel}}} - \underbrace{(m_{^{28}\text{Si}} - 7m_{\alpha})}_Q \quad \text{if } 7\alpha \text{ from } ^{28}\text{Si}^*$$

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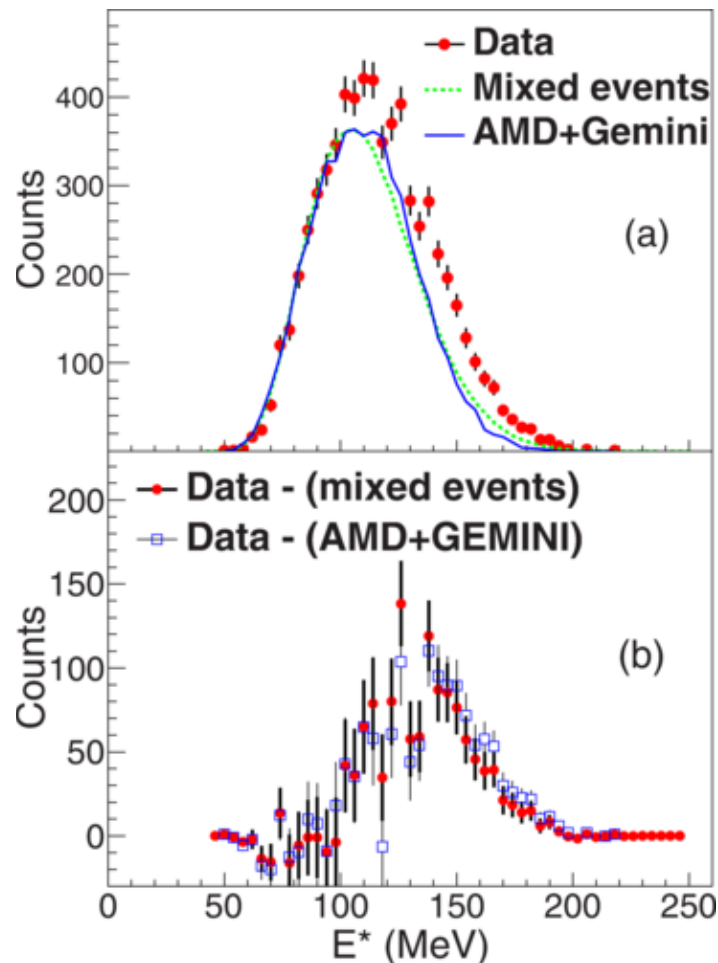
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Neutron Ion Multidetector For Reaction Oriented Dynamics (NIMROD)



First Experimental Evidence (2019)

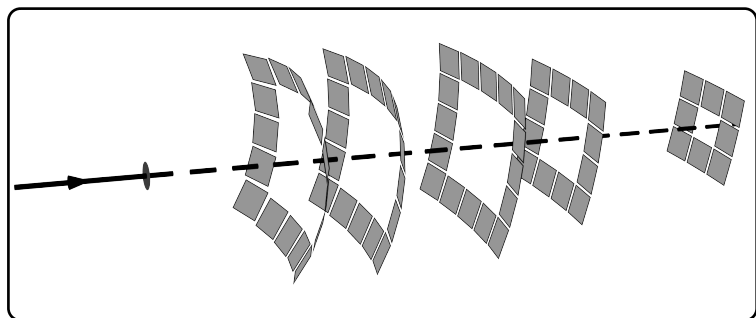
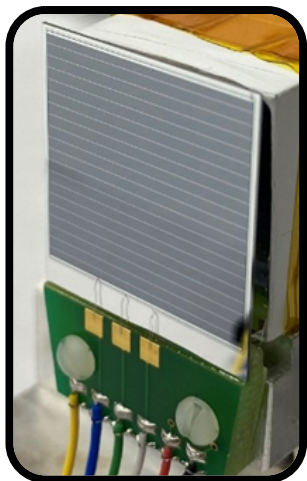
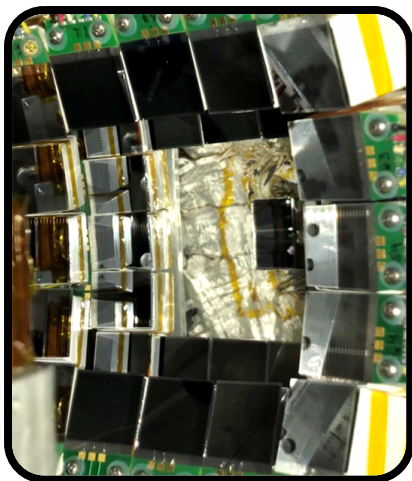
- Reaction
 - $^{28}\text{Si} + ^{12}\text{C}$ @ 35 MeV/u
 - Part of 2009 experimental series
- Sample Size
 - 6,467 Events w/ 7α
- Resolution for E^* from 7α
 - ~ 9.4 MeV (FWHM) in ROI
 - Position insensitive detectors



- Nonresonant Background
 - Mixed Events
 - Shifted AMD distribution
- Extracted Peaks
 - $E^* = 114, 126, \text{ \& } 138$ MeV
 - FWHM dominated by resolution

“Clearly an experiment with much better angular resolution, allowing better resolution for the excitation energy spectrum, will be very desirable.”

Forward Array Using Silicon Technology (FAUST)



Forward Array Using Silicon Technology (FAUST)

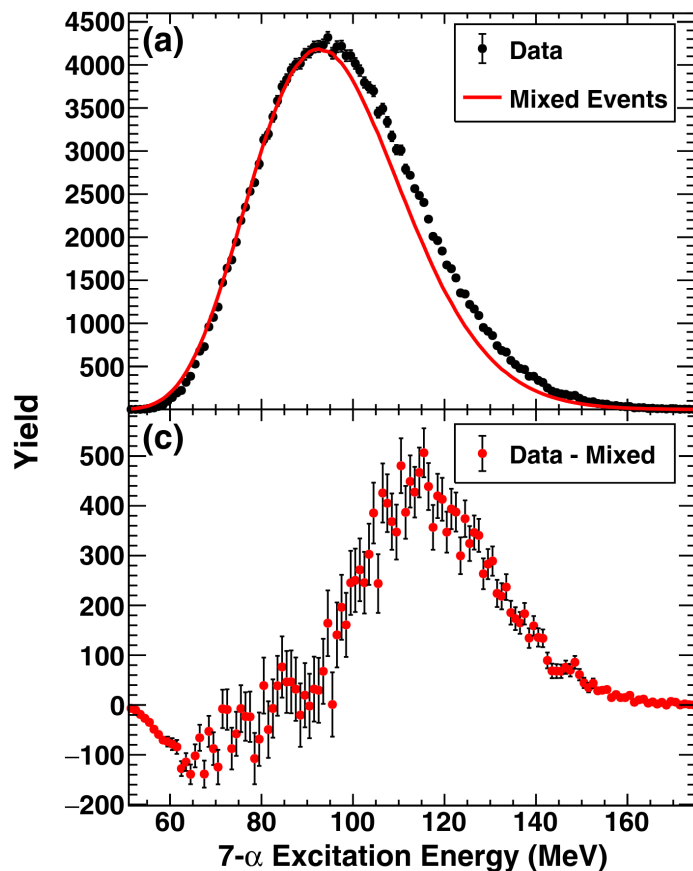
- 68 Si-CsI(Tl) telescopes
 - Excellent Forward Coverage ($1.7^\circ - \sim 40^\circ$)
 - Lab Energy (Thick CsI stops particles)
 - Isotopic Particle Identification ($E-\Delta E$ technique)
 - Lab Angle (Dual-Axis Duo-Lateral detectors)

Measured Reactions

- Target: ^{12}C
- Beam Energy: 35 MeV/u
- Projectiles
 - ^{16}O , ^{20}Ne , ^{24}Mg (Develop analytical techniques + future cluster studies)
 - ^{28}Si (Andy Hannaman's thesis work – confirm and characterize states from Cao et al)
 - ^{32}S , ^{36}Ar (Search for new states predicted by Wong)

Second Experimental Result (2023)

- Reaction
 - $^{28}\text{Si} + ^{12}\text{C}$ @ 35 MeV/u
 - Dedicated measurement for toroidal search
- Sample Size
 - 187,067 Events w/ 7α (x29 $\uparrow$)
- Resolution for E^* from 7α
 - ~ 2.5 MeV (FWHM) in ROI (x3.5 $\downarrow$)

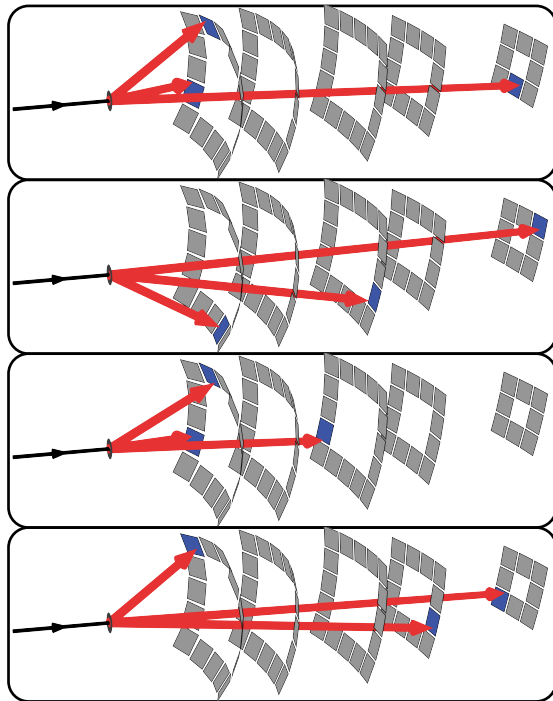


- No obvious peaks in the raw spectrum.
- Backgrounds
 - AMD
 - Model dependence
 - Wrong mean energy
 - Mixed Events
 - Known bias (especially at low energy)

A physically motivated and well benchmarked data driven background estimate is desired for confident interpretation

Creating a Mixed Event

Real Events

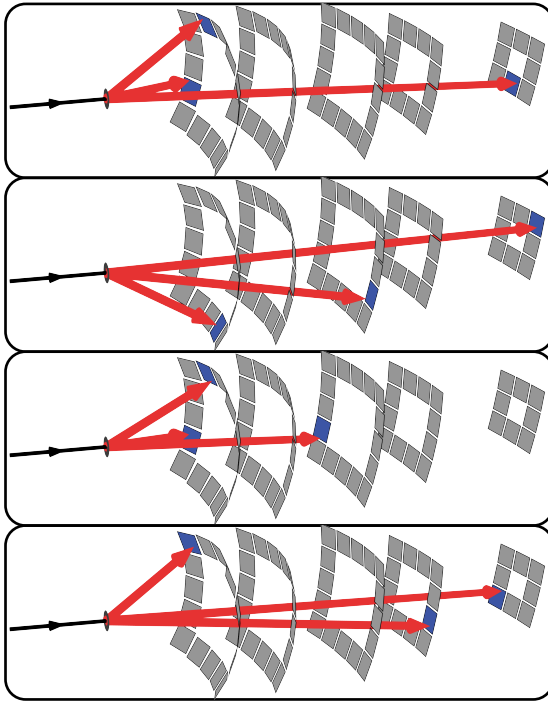


⋮

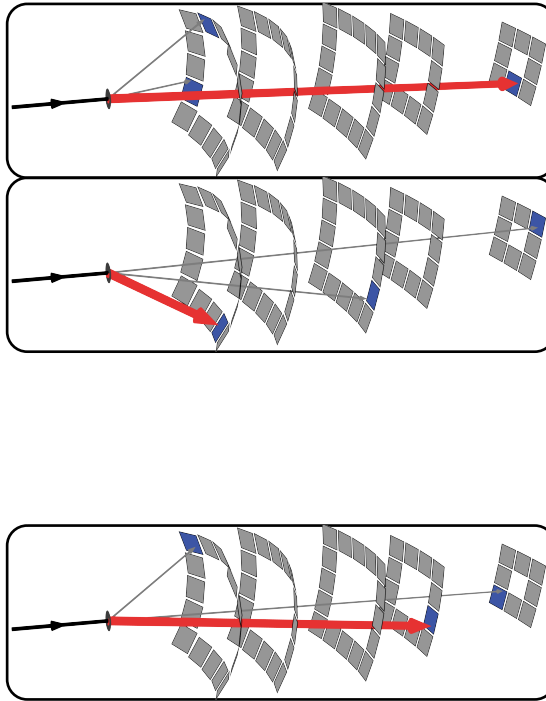
- Start w/ events containing all particles

Creating a Mixed Event

Real Events



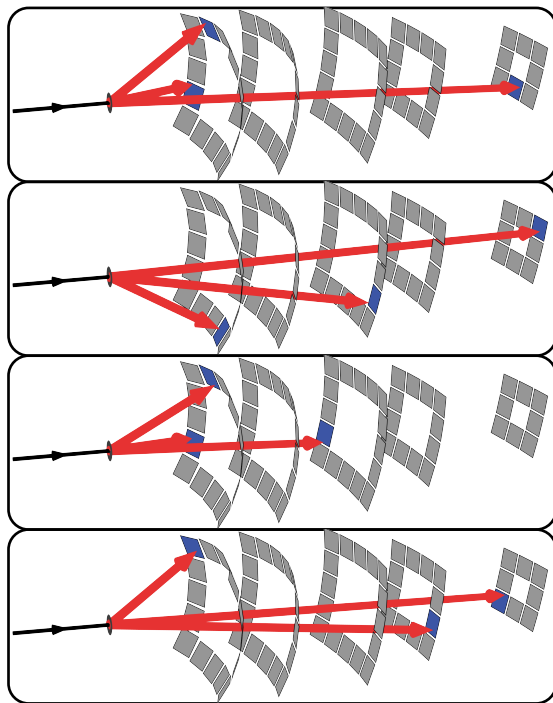
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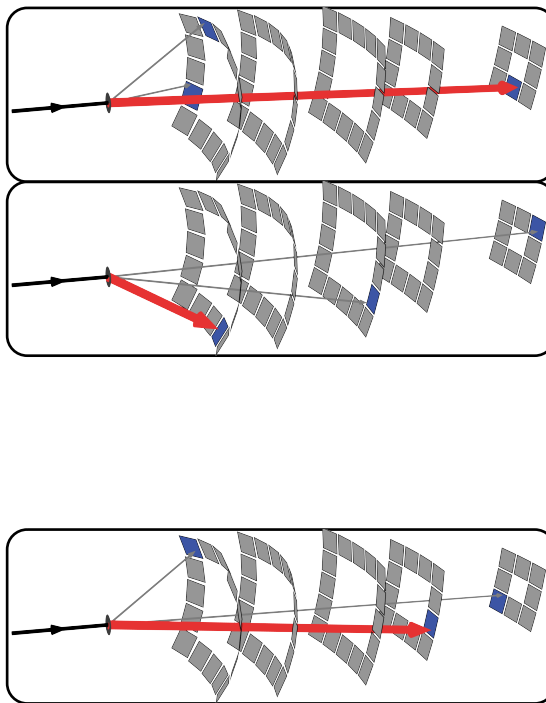
- Start w/ events containing all particles
- Select particles from separate events

Creating a Mixed Event

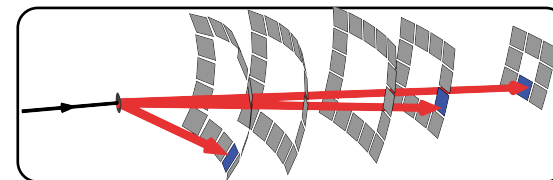
Real Events



⋮



Mixed Events



⋮

- Start w/ events containing all particles
- Select particles from separate events
- Construct new event

Outline

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